Supplemental Material

Are the phases real? Distinguishing bounded interaction groups from spatially structured drift in lower Mississippi Valley decorated ceramics

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Supplement to the main text. Figure and section references to the main text (for example, Figure 4, Results, Discussion) point to the primary article. Citations resolve against references.bib.

Full specification of the four signatures

Departure from neutral transmission follows Neiman's (1995) application of the infinite-alleles model. We estimated the transmission parameter θ in two ways from each assemblage. The homozygosity estimator uses the unbiased sample homozygosity $F = \sum_i n_i(n_i - 1)/[N(N-1)]$, where n_i is the count of class i and N the assemblage size, and inverts the neutral relation $F = 1/(1+\theta)$ to give $\theta_F = (1-F)/F$. The richness estimator solves the Ewens (1972) sampling expectation for the number of classes k,

$$\mathbb{E}[k \mid \theta, N] = \sum_{i=0}^{N-1} \frac{\theta}{\theta + i},$$

for the value θ_E that reproduces the observed k. Under neutral drift, the two estimators agree. Conformist copying concentrates frequency on common classes, raising F and lowering θ_F relative to θ_E, whereas anti-conformist novelty bias does the reverse. The signature is the ratio θ_F/θ_E, which both detects a departure from neutrality and signs it. Because pooling assemblages from divergent groups cancels out the effect, we computed ratios within spatial clusters and averaged the results across clusters.

Seriation-coherence breakdown follows the deterministic frequency-seriation method of Lipo, Madsen, and Dunnell (Cowgill 1972; Dunnell 1970; Graves and Cachola-Abad 1996; Lipo et al. 1997, 2015; Lyman and O'Brien 2006; Marquardt 1978; O'Brien and Lyman 1999). A set of assemblages is co-seriable if some ordering renders every decorated class's frequency curve single-peaked and without gaps, the battleship-shaped distribution of frequency seriation. We enumerated the complete set of maximal co-seriable groups, the subsets for which a valid ordering exists, and which are maximal under inclusion. A single group indicates one interacting community, whereas a proliferation of small, mutually non-co-seriable groups indicates a fragmented one. We report the number and maximum size of these groups and, for each assemblage, its group-membership count, which identifies the bridge assemblages that join otherwise incompatible orderings. The number of maximal groups is the continuous signature.

The variance partition is a cultural analog of Wright's F_{ST} computed on Gini-Simpson diversity $H = 1 - \sum_i p_i^2$,

$$F_{ST} = \frac{H_T - H_S}{H_T},$$

where H_T is the diversity of the pooled assemblages and H_S the size-weighted mean within-cluster diversity (Bell et al. 2009). It is zero when clusters share a repertoire and rises as they come to hold distinct ones.

Spatial assortativity asks whether ceramic similarity follows social rather than geographic distance. We measured Brainerd-Robinson similarity $S_{ij} = 200 - \sum_k |100\, p_{ik} - 100\, p_{jk}|$ between every pair of assemblages. Because a Mantel correlation cannot separate a sharp boundary from a smooth distance decay, we summarized assortativity as a distance-controlled boundary excess. Within each pairwise-distance bin, the mean within-cluster similarity minus the mean between-cluster similarity is averaged over bins. Under pure isolation by distance, similarity depends only on distance, and the excess is near zero. A sharp social boundary makes within-cluster pairs much more similar than between-cluster pairs at matched distance, yielding a large positive excess. Spatial clusters were obtained by k-means on site coordinates with multiple restarts, the number of clusters chosen by the mean silhouette.

We combined the four signatures by standardizing each across the ordinal sequence and averaging them, so that no single signature would dominate, and summarized the trajectory as the ordinal slope of the combined score and each component, with rank correlations and bootstrap confidence intervals.

The generative validation model

The validation rests on a generative model in which decorated variants are copied under neutral transmission (Bentley et al. 2004; Neiman 1995), with three mechanism-faithful controls: the between-group divergence of class profiles, the strength of within-group conformity (a frequency-dependent copying bias, after (Henrich 2001)), and a spatial rule placing group profiles on the landscape. Four generators instantiate the four idealized processes of Table 1, where drift is a single process that the main-text table shows under two regimes, smooth and stochastic, of which the smooth, deterministic regime is the one generated here. The bounded-groups generator raises divergence and conformity together across spatially bounded groups. Environmental patchiness holds a fixed between-group difference under neutral within-group copying with an environment-keyed layout. Aggregated conformity raises within-group conformity in a single undivided pool with no between-group divergence. Drift with distance-limited interaction produces a smooth spatial gradient under neutral copying. Each generator emits a sequence of group-by-class slices along an ordinal axis. We ran the four-signature procedure on each generator, blind to its source, and recorded which registered as convergent, repeating this across 500 random seeds. On the bounded-groups output we also computed the correlation matrix of the four signatures, to confirm that their joint movement is informative rather than an artifact of redundancy (Figure S1). This idealized test uses large, clean assemblages (12 groups, several hundred sherds each) and establishes that it discriminates in principle. The record-matched, size-controlled recovery in the main text (Figure 4) is what establishes which signatures survive at the resolution of the actual data.

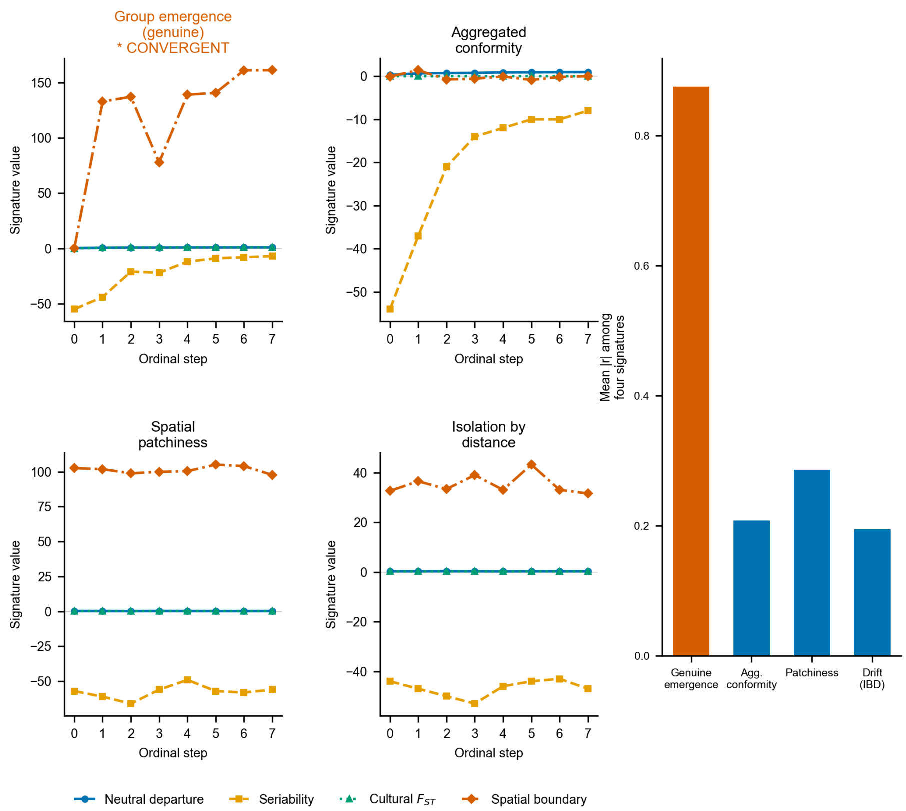

Figure S1. Idealized criterion validation. Four-signature ordinal trajectories under each of the four generative mechanisms on large, clean synthetic assemblages. Only a bounded interaction group drives all four together (starred), and no mimic is flagged as convergent across 500 random seeds (the bounded-groups case flagged in 493 of 500, 95 percent interval 0.97 to 0.99, and no mimic flagged at all, 95 percent upper bound about 0.006). Inset: the signature-independence audit (mean absolute pairwise correlation among the four signatures), high under closure (about 0.88), and low under the mimics (about 0.2). This shows the convergence concept works on idealized data. Figure 4 shows what survives at the record's resolution.

Robustness checks

Neiman's (1995) sample-size diagnostic for the neutrality estimators supports the Discussion. Across the 29 basin assemblages, we computed the homozygosity-based and richness-based estimates of the transmission parameter and their signed difference. We then correlated each with assemblage size and with seriation position, following Neiman's protocol for separating a transmission signal from a sample-size artifact. Several further checks support the Results. A forking-paths sensitivity grid recomputes the convergence verdict across twenty-seven combinations of the basin's spatial cutoff, bin count, and cluster number. The frequency increment test (Feder et al. 2014) is applied to each decorated class's binned frequency trajectory. Neiman's (1995) squared-Euclidean interassemblage-distance is computed with its diversity-distance relationship and divergence trajectory. The alternative basin definition, the stricter watershed rule, is evaluated in full. Clustering-tendency statistics (Hopkins, silhouette, and the gap statistic) (Hopkins and Skellam 1954; Rousseeuw 1987; Tibshirani et al. 2001) on the correspondence-analysis ordination favor a single group, with a maximum silhouette of only 0.47, consistent with a near-continuum rather than sharply bounded ceramic groups. All are provided in the supplementary code. The Neiman interassemblage-distance result is shown in Figure S2.

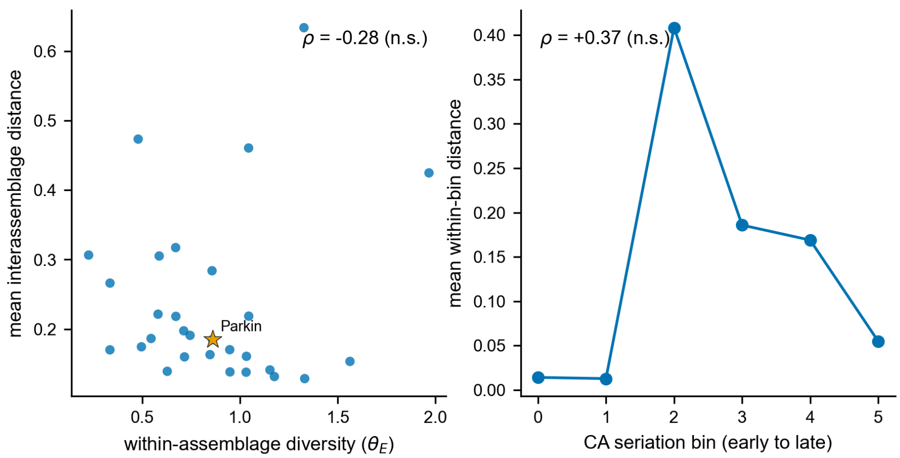

Figure S2. Neiman interassemblage-distance result. (A) Within-assemblage diversity against mean interassemblage distance, with Parkin starred. The weak negative association (Spearman rho = -0.28, not significant) is directionally consistent with the drift signature Neiman (1995) reported at rho = -0.62, not with the positive relationship that bounded, assorting groups would produce. (B) Mean within-bin interassemblage distance along the CA seriation axis. Between-assemblage divergence does not grow toward contact (rho = +0.37, not significant).

The F_{ST}-led reduction is a property of the record, not the generator

The recovery experiment injects group closure as between-cluster divergence, which is what cultural F_{ST} measures, so one objection is that the experiment only certifies an F_{ST} test recovering an F_{ST}-shaped signal. We tested this by injecting closure through two further pathways, each matched to a different signature, and asking which signature recovers each. A within-group conformity pathway raises conformity in a single undivided pool with no between-group divergence, the pathway that the neutral departure is meant to detect. A non-spatial social pathway assembles assemblages into social groups scrambled relative to the spatial clusters, the pathway seriation coherence is meant to detect. The result is one-sided in two ways. First, the conformity pathway is recovered by no signature (the neutral departure, which should detect it, inverts rather than rises), and the non-spatial social pathway is not recovered by seriation either. Hence, the neutral and seriation signatures fail to register an injected signal even through the pathway each is designed for. Second, F_{ST} is not a clean spatial-divergence test but a near-omnibus one, responding to group structure, however it is organized. It recovers the spatial-divergence pathway strongly (mean trend rising from the null to +0.76 at strong injection) and also recovers the non-spatial social pathway (+0.77), because both ultimately raise between-group variance somewhere the partition can see it. The four signatures are therefore not four independent pathways, even in this diagnostic. The only one that registers an injected signal at this resolution is F_{ST}, and it does so however the underlying divergence is organized. The reduction to F_{ST} is a property of the record's resolution, not an artifact of a generator whose injected closure is collinear with F_{ST}. That resolution is 29 time-averaged assemblages subsampled to a common count over ten aggregated classes. A subsampling-depth sweep (the weak signatures climb modestly with more sherds but remain well below F_{ST}) and a cluster-count sweep (the spatial signature stays weak through k = 5) confirm that the limitation is the record's size and time-averaging rather than a single tuning choice. The consequence, stated in the Discussion, is that the negative bounds only cover closure expressed as between-group divergence. A closure carried purely by conformity or by a non-spatially organized social structure is below detection here at any strength.

Positive control: the basin matches neutral spatial drift

To test what process the basin's structure matches, rather than only which process it fails to show, we simulated neutral cultural drift on the real coordinate layout of the 29 assemblages, following the finite-population network-drift model of Lipo, DiNapoli, Madsen, and Hunt (2021). Each assemblage is a node holding a finite population of learners that copies decorated-class variants within the node and, at a low rate, from other nodes with a probability that decays with geographic distance, with innovation and the record's trailing-window time-averaging imposed. We compared this neutral model against a bounded-groups model in which interactions are concentrated within spatially contiguous groups, with only a small leak between them, both run on the same coordinates and subsampled to the observed per-assemblage counts. We swept the interaction range (6, 12, and 24 km), the between-node mixing rate (0.02, 0.05, and 0.20), and the innovation rate (0.005, 0.01, and 0.02), in both snapshot and time-averaged form, at 100 seeds per cell. For each cell we recorded whether the four observed statistics fall inside the neutral null. Across the time-averaged sweep, the region that brackets all four at once is the low-mixing, sub-basin-range corner (24 km, mixing 0.02, innovation 0.01), as reported in Figure S3. At shorter ranges, drift over-produces the distance decay. At higher mixing, it underproduces the structure, so the all-four match holds only in a plausible but specific region of parameter space rather than across the whole parameter space. In every cell, the bounded-groups model overshoots the three structural statistics by roughly two to seven times relative to the observed values. The contrast between drift, which can reproduce the data, and bounded groups, which cannot, is therefore robust even where the all-four drift match is not. The control also demonstrates that the distance-controlled boundary excess, near zero under smooth deterministic isolation by distance, takes substantial positive values under stochastic neutral drift, so a positive boundary excess does not by itself distinguish a social boundary from drift on a finite network.

The time-aware drift simulation

The forward demonstrations in the main text (Figures 8 and 9) and the basin positive control (Figure S4) use the same neutral-drift model run on the real geography, distinct from the null-comparison of the positive control above. The model is run at two scales: on the 29 St. Francis basin assemblages for the positive control (Figure S4) and on all 55 Mainfort-PFG decorated assemblages of the wider lower Mississippi Valley for the region-wide group counts (Figure 8) and the Parkin pull-out (Figure 9). The model evolves a population of decorated-class variants at each assemblage location. Each location holds a fixed finite population of 120 learners. In each generation every learner copies the decorated class of another learner. That partner is drawn from the learner's own location with high probability and, with probability 0.02, from another location. The other location is chosen with a weight that decays exponentially with along-waterway (river-network) distance, described below, at a length scale of 24 river-km. With probability 0.012 a learner instead adopts a newly invented class, so the decorated repertoire turns over and new classes appear through time. That turnover is the source of the differential late-class frequencies that let assemblages be seriated. No interaction boundary is imposed anywhere, so any structure in the output is produced by river-network position and time alone. The engine is the finite-population network-drift model of Lipo, DiNapoli, Madsen, and Hunt (2021), here run forward with innovation switched on rather than set to zero to isolate structure.

Interaction in this riverine settlement system travels along watercourses, so the

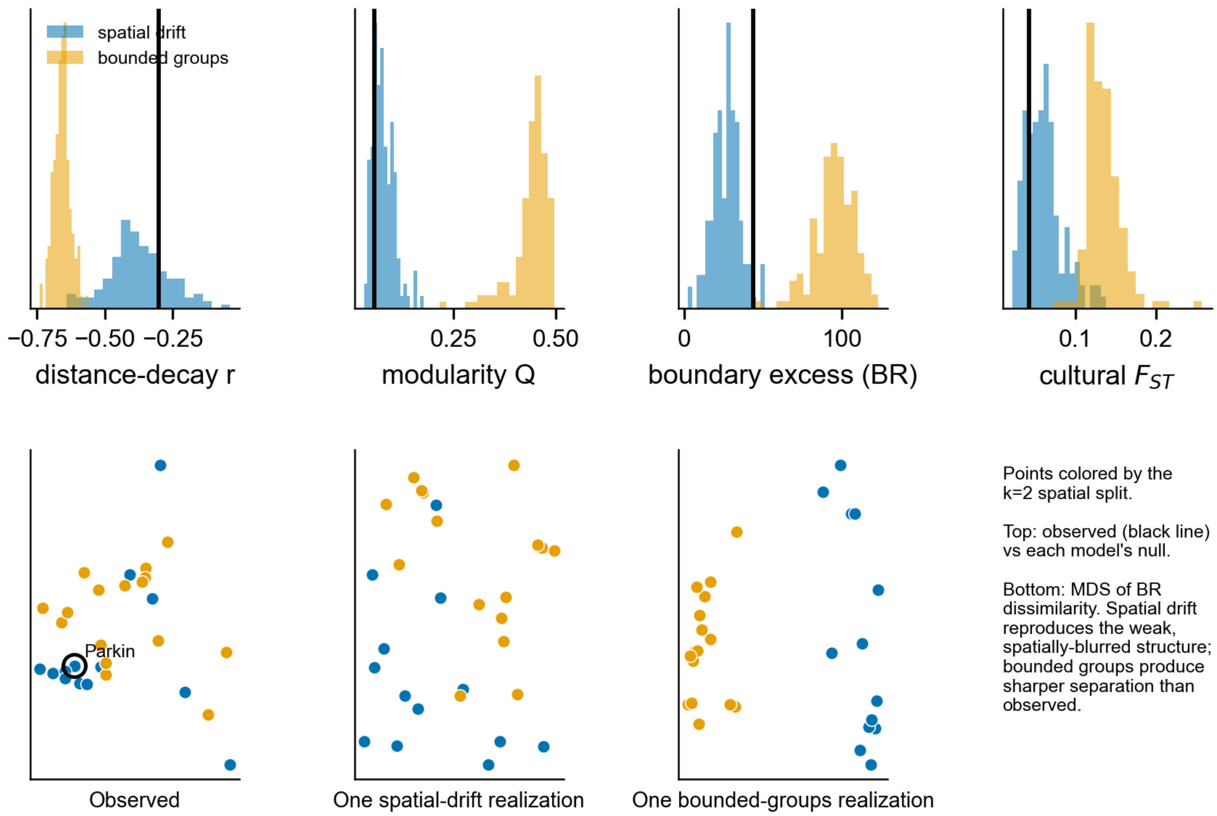

Figure S3. Spatial drift versus bounded groups on the basin geography. Top: the four observed signatures (vertical lines) against the null distributions produced by a neutral spatial-drift model (blue) and a bounded-groups model (orange) on the real coordinate layout, both time-averaged and subsampled to the observed counts. Observed values fall inside the drift distribution and far below the bounded-groups distribution for modularity, boundary excess, and cultural F_{ST}. Bottom: multidimensional scaling of decorated dissimilarity for the observed assemblages and one realization of each model, spatial drift reproducing the weak, spatially blurred structure while bounded groups produce a sharper split than the data show.

copying kernel uses river-network distance rather than straight-line distance. We build a graph from the mapped hydrography (the LMVHydrology centerlines clipped to the basin with a 25 km margin), repair its topology by bridging vertices within 250 m of one another, keep the largest connected component, and snap each assemblage to the nearest network node, adding the snap distance as an access cost at both ends. The distance between two assemblages is the shortest along-water path between their snapped nodes. All 29 assemblages connect on the resulting network, the mean access distance from an assemblage to a mapped channel is 3.1 km, and along-water paths average about 2.3 times the straight-line distance. Substituting straight-line distance changes the results negligibly (between-group F_{ST} and community counts shift within sampling error), so the conclusions do not depend on the choice. We use river-network distance because it more faithfully represents how potters in the basin could interact. The same construction over all 55 lower-valley assemblages yields a single connected network for the wider-valley analyses (maximum access 19.6 km).

The assemblages are not contemporaneous, which is both why they can be seriated and why some carry more of the late classes than others, so the evolving field is sampled time-transgressively rather than at a single instant. The field is recorded at 90 evenly spaced time slices over 1,800 generations. Each synthetic assemblage is drawn from the slice that matches its observed seriation position, its rank on the first correspondence-analysis axis of the real data. It is accumulated over a trailing window of eight slices so that, like a real death assemblage, it mixes production across its span. Each synthetic assemblage is then sampled to the observed sherd count of the corresponding real assemblage. The synthetic assemblages are treated exactly as the empirical ones: interaction communities by the same modularity partition, between-group cultural F_{ST} on the resulting groups, and a seriation order from a fresh correspondence analysis. A contemporaneous control, with every assemblage drawn from the final slice, is sampled the same way for comparison.

For the basin positive control (Figure S4) we summarize 500 independent realizations at the positive-control corner (interaction length 24 river-km, between-node mixing 0.02, innovation 0.012). Because group labels are arbitrary and switch between runs, groupness is summarized by co-membership rather than by a single partition. For each assemblage, this is the fraction of the 500 runs in which it falls in the same drift-detected group as Parkin. This quantity is label-invariant, needs no alignment across runs, and expresses the contingency of any single solution as a probability surface. Figure S4A shows this for the contemporaneous snapshot, in which every assemblage is drawn from the same final time slice, so it isolates the spatial signal. Co-membership with Parkin falls off with river distance. Under the full time-transgressive sampling that gradient disappears, because assemblages drawn at different seriation positions carry different repertoires and fall into different, time-defined groups. Membership then reflects space and time jointly, and a site's nearest neighbors need not share its phase. Either way, no assemblage is a certain member or non-member, so the phase has no stable membership. Figure S5 repeats the experiment across a grid of the three governing parameters to show that the result does not depend on this corner. The same model applied to the wider 55-assemblage valley gives the region-wide group counts of Figure 8 and the Parkin pull-out of Figure 9.

Robustness of the drift-generated structure

The positive control shows the observed structure falls inside the drift distribution. The wider-valley demonstration (Figure 8) goes further, generating phase-like structure from drift on the real geography, and we checked that this result is not contingent on a single stochastic draw or a tuned parameter set. We repeated the time-transgressive drift experiment across 432 runs spanning a grid of interaction length (12, 18, 24, and 36 river-km), between-node mixing (0.01, 0.02, and 0.05), and innovation rate (0.006, 0.012, and 0.024), with twelve seeds per cell (Figure S5). Two or more spatially coherent groups, the phase-like territories a culture historian would name, appear in 431 of the 432 runs. The between-group cultural F_{ST} sits at or below the observed value in 85 percent of runs and within a factor of two of it in 53 percent, short of the bounded-groups expectation throughout. F_{ST} is essentially flat across interaction length and declines with more mixing or more innovation, the expected behavior of a drift process. The synthetic record re-seriates across the full grid (mean absolute Spearman correlation 0.44 between recovered and imposed order). The appearance of phase-like structure at the observed magnitude is therefore a generic outcome of distance-structured drift on this geography rather than a product of the chosen parameters.

Regional extension: the southeast-Missouri phases

We extended the within-region test to the southeast-Missouri survey assemblages of Williams (1954), comprising 50 assemblages from 28 sites, digitized from the survey tables and georeferenced using the Lower Mississippi Survey grid. Williams divided these lowlands into four named Mississippian phases. Across the named physiographic regions the cultural F_{ST} is 0.11. Most of that differentiation is a Woodland-to-Mississippian composition gradient rather than synchronic social structure. Restricting the analysis to Mississippian classes collapses the F_{ST} to 0.03, near the drift level. Within the region the assemblages do not separate into bounded clusters. They intermix on the ordination, and the between-cluster F_{ST} sits at the neutral-drift expectation and far below the bounded-groups null (Figure 11). Williams's southeast-Missouri phases, like the named phases of the St. Francis basin, are arbitrary divisions of a continuous field structured by geography and time. This is the same verdict Fox (1998) reached for these phases by cluster analysis and ordination.

Dynamic-sufficiency tests in the time domain

The four convergence signatures are measured as states along the seriation sequence, so on their own they are only partially dynamically sufficient. They

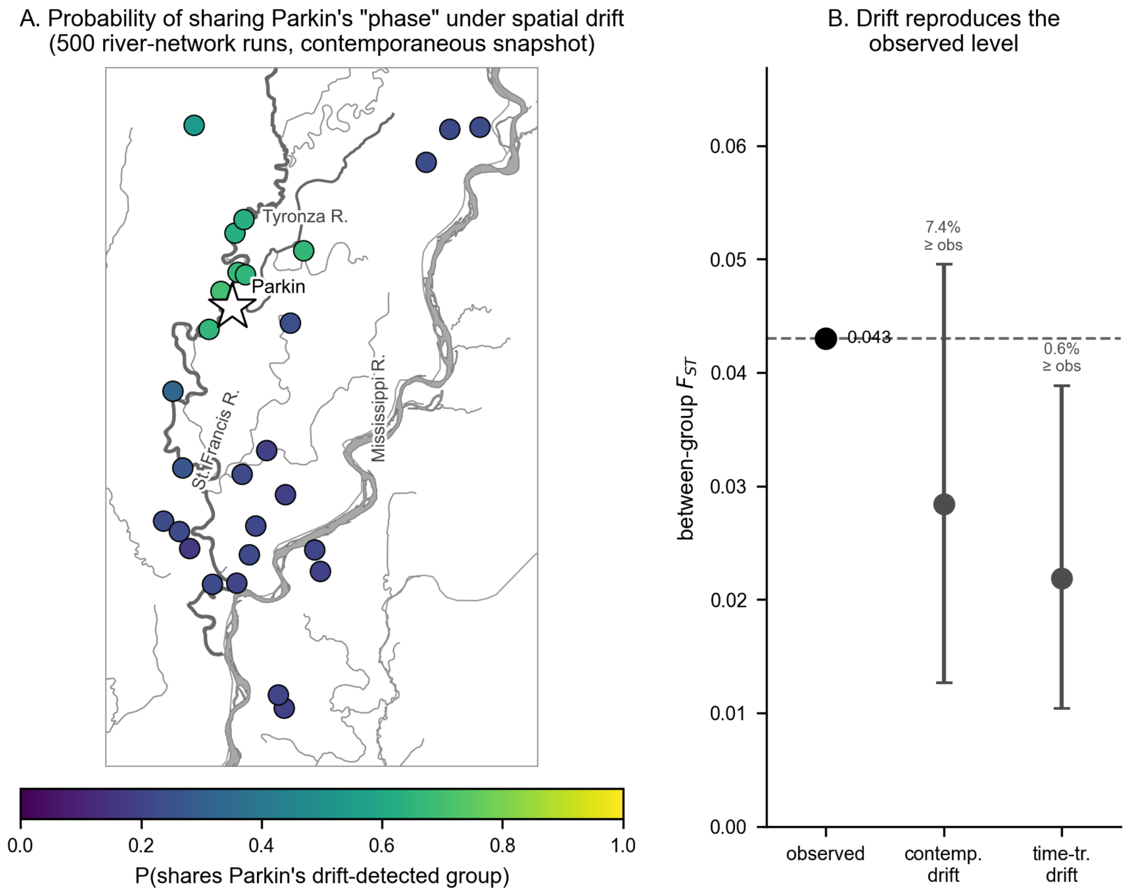

Figure S4. Basin positive control: phase-like structure generated by neutral drift on the 29 St. Francis basin assemblages, with no bounded group imposed, over 500 realizations. (A) Each assemblage is shaded by the probability that it shares Parkin's drift-detected group (star) under the contemporaneous snapshot that isolates the spatial signal. The probability falls off smoothly with river distance from Parkin (Spearman rho = -0.72), from about 0.70 nearby to 0.17 far away, so Parkin's "phase" is a graded, probabilistic field rather than a bounded set. (B) Between-group cultural F_{ST} for the observed data (0.04) against contemporaneous and time-transgressive drift (medians with 95% intervals across the 500 runs). The observed value falls within the contemporaneous-drift interval, with 7.4% of drift runs reaching or exceeding it, so drift reproduces the observed level rather than the higher value a bounded group would leave.

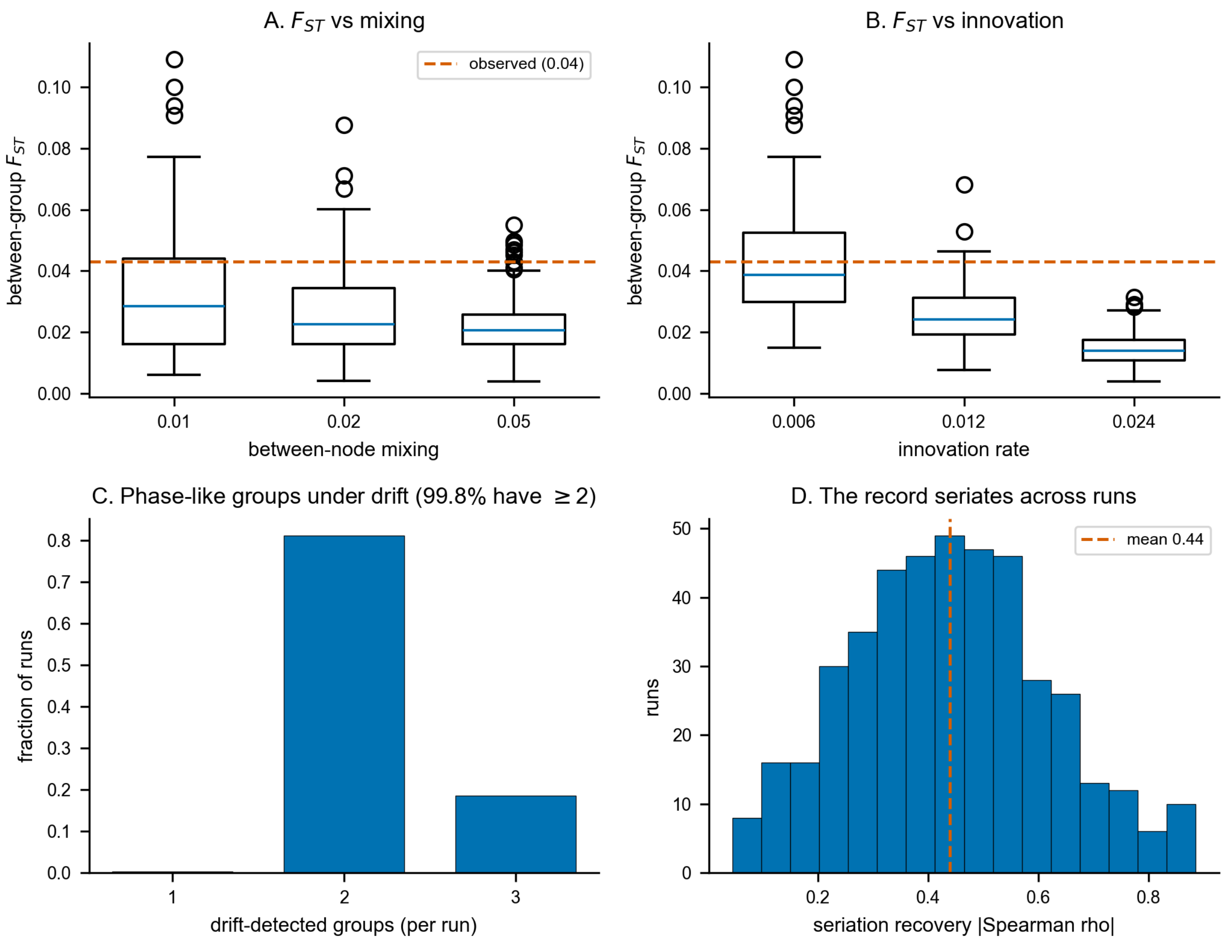

Figure S5. Robustness of phase-like grouping to contingency and model factors, across 432 time-transgressive runs (a grid of interaction length, between-node mixing, and innovation rate, twelve seeds per cell). (A) Between-group cultural F_{ST} by between-node mixing and (B) by innovation rate, boxes taken over all other factors and seeds. The observed value (dashed) is bracketed, and F_{ST} falls with both factors but never reaches the bounded-groups level. (C) Distribution of the number of drift-detected groups per run, with 431 of the 432 runs (99.8 percent) yielding two or more. (D) Distribution of seriation recovery, the absolute Spearman correlation between the recovered CA1 rank and the imposed time order, across runs (mean 0.44).

describe states rather than the transitions between them. Two further tests work on the transformation between states, and a third, underpowered probe is reported in the code for completeness. The two we rely on are dynamically and empirically sufficient at once, because each models the change between states while respecting the time-averaged resolution of the record.

The first infers how potters were copying by simulation. We run many copying scenarios, ranging from strongly conformist to neutral to novelty-seeking, and keep only the ones that reproduce the real assemblages. From the surviving scenarios we infer the kind of copying implied, with the record's time-averaging built into each simulation (the approximate-Bayesian-computation approach) (Carrignon et al. 2019; Crema et al. 2014, 2016). We modeled the basin sequence as frequency-dependent copying among the ten decorated classes in a finite population. Each generation, the adoption weight of a class is proportional to its frequency raised to the power $1 + b$, where $b > 0$ is conformity, $b < 0$ is anti-conformity, and $b = 0$ is neutral copying, with an innovation rate that mixes in a uniform draw from the set of classes. An assemblage pools production over a window of generations, so the time-averaging that distorts the static statistics is built into the generative process rather than corrected after the fact. This is what makes the test both empirically and dynamically sufficient. We placed broad priors on the innovation rate, the bias b, the population size, and the time-averaging window, and fit the model by sequential Monte Carlo approximate Bayesian computation (800 particles advanced through six rounds of shrinking tolerance), which advances a population of parameter sets through a sequence of shrinking distance tolerances and corrects the accepted values with a local-linear regression adjustment, matching each simulated six-bin sequence to the observed diversity trajectory and rank-frequency profile (Beaumont et al. 2002; Toni et al. 2009). The posterior of the bias is tightly constrained near neutral, with mean $b = +0.006$ and a 95 percent interval of -0.020 to $+0.035$ that includes zero (Figure S6A). The posterior standard deviation is 0.014 against a prior of 0.289, so the data strongly constrain the bias rather than leaving it at the prior. Any conformist lean is weak and unresolved, with $P(b > 0) = 0.70$, and refitting the bias separately to the early and late halves of the sequence recovers no shift between them. The inference rules out a strong basin-wide conformist bias by modeling copying dynamics rather than a single static moment. Because it fits the sequence as a single pooled population, it bears on aggregate conformity across the basin rather than on a spatially localized or bounded-group version, which the spatial signatures and the drift comparison address. We validated the method by simulation-based calibration, fitting 200 datasets drawn from the prior and confirming that the rank of each true value within its posterior is uniform (Figure S6B) (Talts et al. 2018). The calibrated posterior is concordant with an independent rejection-sampling fit of the same model (mean $b = +0.028$, a wider 95 percent interval that also includes zero), which the calibration sharpens rather than overturns. Full configuration, diagnostics, and software versions are in the supplementary code.

The second test asks, in plain terms, whether the assemblages' diversity drifts without direction through time, heads steadily in one direction, or is pulled back toward a stable value, by comparing four standard time-series models (an unbiased random walk, a directional walk, stasis, and a mean-reverting Ornstein-Uhlenbeck process) (Hunt 2006; Hunt et al. 2008). For the Gini-Simpson diversity trajectory and for the abundant decorated classes, binned along the oriented axis, we fit four time-series models on the same level vector: an unbiased random walk (Brownian motion, the neutral-drift expectation), a general biased random walk (the directional signature a sustained transition would leave), a stasis model (white noise about a constant mean), and an Ornstein-Uhlenbeck model (mean reversion toward an optimum). We compared them by the small-sample Akaike information criterion. The aggregate diversity trajectory favors the unbiased random walk, with an Akaike weight of 0.81 against 0.19 for the directional model and negligible weight for stasis or the Ornstein-Uhlenbeck model. The model comparison has only six points, so it is weak. This result is a concordance rather than a decisive selection (Figure S6C). The diversity trajectory is the trait of interest here, because the correspondence-analysis ordering does not arrange it by construction. Individual-class trajectories more often favor the directional model. That comparison, however, is weak and partly tautological because the seriation axis is designed to maximize monotonic class-frequency turnover, and only three classes are abundant enough across the sequence to fit. The robust conclusion is neutral drift, not a sustained directional transition.

A third probe, for the early-warning signature of a critical transition, is underpowered at this ordinal resolution and does not meet empirical sufficiency. Across the correspondence-analysis-ordered sequence the rolling variance shows no rising trend (Kendall tau $= -0.03$) and the lone rise in lag-1 autocorrelation (tau $= +0.78$) is an artifact of the ordering, which places compositionally similar assemblages adjacent, so we report it in the supplementary code but do not rely on it (Dakos et al. 2012; Scheffer et al. 2009).

Radiocarbon chronology

We calibrated the Mainfort (2001) radiocarbon compilation against IntCal20 (Reimer et al. 2020). Forty-one determinations come from seven St. Francis Basin Parkin-phase sites, nineteen of them from Parkin. The summed probability distribution of the basin dates concentrates the occupation in the fourteenth to sixteenth centuries, with a median of AD 1422 (Parkin alone, AD 1483). It declines across the contact interval: 29 percent of the basin probability mass postdates AD 1541 and 18 percent postdates AD 1600 (Figure S7). This is consistent with occupation continuing through the De Soto entrada, which the chronicles place at Casqui, and declining thereafter, rather than with a sequence running unbroken into the seventeenth century. However, the protohistoric tail is partly a calibration artifact of low-activity determinations near the post-1650 plateau (Surovell et al. 2009; Surovell and Brantingham 2007). To test the assumption that the seriation axis tracks calendar time, we pooled the dates of the five proveniences matching curated assemblages. We correlated their pooled

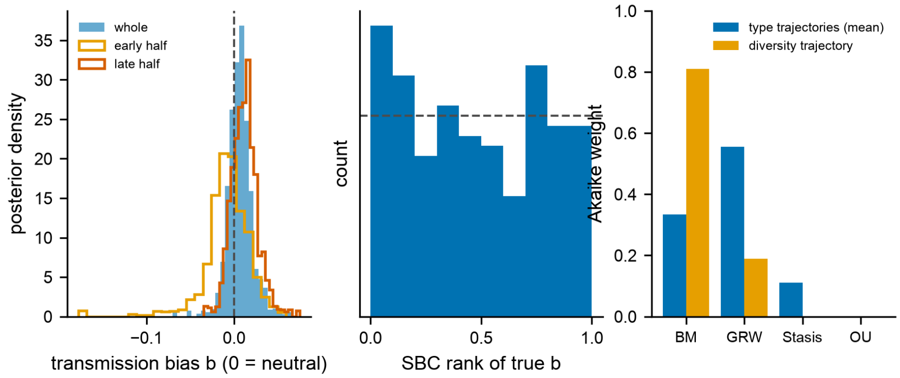

Figure S6. Dynamic-sufficiency tests. (A) Posterior of the transmission-bias parameter b from the generative, time-averaging-aware inference, fit by sequential Monte Carlo approximate Bayesian computation ($b = 0$ neutral, $b > 0$ conformist, $b < 0$ anti-conformist). The whole-sequence posterior (filled) and the early- and late-half posteriors (outlines) are all tightly constrained near neutral and include zero, ruling out the strong conformity a marker dynamic would require. (B) Simulation-based calibration of that inference. Over many datasets drawn from the prior, the rank of the true bias within its posterior is uniform (dashed line the uniform expectation), so the posterior is calibrated rather than merely contracted. (C) Tempo-and-mode Akaike weights of the four time-series models (BM unbiased random walk, GRW directional walk, Stasis, OU mean reversion) for the Gini-Simpson diversity trajectory and averaged across the abundant class trajectories. The diversity trajectory, which the correspondence-analysis ordering does not arrange by construction, favors the unbiased random walk (neutral drift), a concordance rather than a decisive selection given only six points.

median calibrated ages with CA1 position: Spearman's r = +0.70 (n = 5, 27 dates pooled), with four sites in calendar order and one late-dated site, Clay Hill, departing from it. The orientation of the axis toward later time is supported but not certain, and we treat directional statements as provisional throughout.

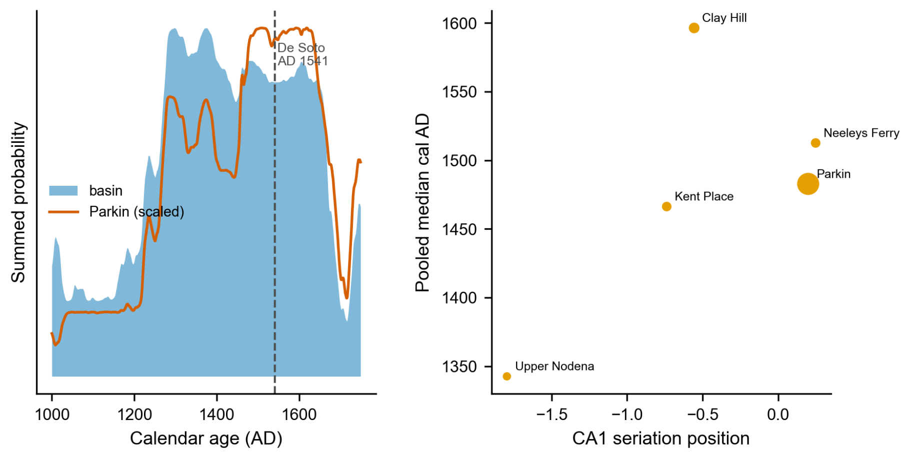

Figure S7. Radiocarbon chronology of the St. Francis basin (Mainfort determinations calibrated against IntCal20). Left: summed probability distribution for the basin Parkin-phase sites (filled) and for Parkin alone (line, scaled), with the De Soto entrada (AD 1541) marked. Occupation concentrates in the fourteenth to sixteenth centuries and declines across contact. Right: CA1 seriation position against the pooled median calibrated age for the five proveniences matching curated assemblages (point size proportional to the number of dates). The axis tracks calendar age (Spearman +0.70) except for the late-dated Clay Hill.

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